

# Annex XV — Physical-World Object Designation

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**Document status:** Normative unless explicitly marked as informative.

**Applies to:** the spatial embodiment of Scoped Worlds (Annex XIV) — combined entry capture and spatial registration of a device inside an existing Virtual Replica; headless object designation through sight-field recognition, named addressing, gaze, pointing, and proximity; the Spatial Anchor Registry and Pose Authority; identity facets and visible confirmation; geofencing as a context layer; and the binding of WR Advance Preparation and WR Predictive Preparation views to designated physical objects.

**Trademark notice:** Optirando™, WR Desk™, WR Graph™, BEAP™, PoAE™, PoAC™, WRVault™, WR Guard™, and WR Code® are used throughout as product and protocol designations of the Optirando™ system.

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## XV.0 Purpose and Amendment Discipline

This annex specifies how physical objects — machines, robots, autonomous systems, products, tools, documents, and other uniquely addressable physical entities — are entered into, designated within, and acted upon inside a governed WR environment. It formalizes three connected mechanisms:

1. **Floor Entry** — the combined entry capture that opens a spatially anchored Scope, performs the mandatory WR Handshake of Scope Entry, and spatially registers the device inside an existing Virtual Replica;
2. **Headless Object Designation** — the resolution of an individual object's WR identity without a per-object physical scan, using sight-field recognition, spoken names, pointing, gaze, NFC/proximity, WR Pointer selection, or combinations thereof, all inside the established WR Handshake context;
3. **Preparation Views** — the immediate availability of the object's governed information space, including WR Advance Preparation and WR Predictive Preparation briefings.

Amendments to this annex follow the Canon transparency rule: versioned updates only, never silent replacement. Changes to normative principles are permitted when errors or improvements are identified, and must be explicit, versioned, and accompanied by a stated rationale.

### XV.0.1 Claims Discipline (Normative)

The claims discipline of Annex IX §IX.0.1 applies verbatim. No novelty claim is made for any individual mechanism described here. Fiducial-marker tracking, pointing interfaces, multimodal disambiguation, indoor localization, and digital environment twins each have substantial prior work. The contribution disclosed by this annex is the **combined, governed chain**: physical designation →

identity resolution → cryptographic trust context → Virtual Replica → permissions → object-specific action and preparation — with every consequential transition gated by the WR governance model. This combined disclosure is published as prior art. Specific external references discussed during drafting are not cited pending independent verification of publication date, technical scope, and differences.

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## XV.1 Position Within the Canon and the Main Paper

### XV.1.1 Relationship to the Main Paper (Informative)

The main paper already discloses the spatial foundation this annex builds on: the roof/leaf WR Code® hierarchy, in which a roof WR Code® defines the boundaries, geometry, and semantic map of an environment; the co-location of a fiducial marker (e.g., AprilTag) with a roof WR Code® for physical–digital alignment; the principle that a single roof scan loads all subordinate WR Code®s within the spatial group; robot self-localization via fiducial markers inside the WR-Code®-defined coordinate system; geofencing; and 3D environment replicas usable for simulation and robotic training. This annex does **not** re-invent that foundation. It refines it, separates its components by role, and connects them to the Scoped Worlds machinery and the new designation and preparation workflows.

### XV.1.2 Relationship to Annex XIV — Scoped Worlds (Normative)

A **Floor is a spatially anchored Scope**. Everything Annex XIV specifies for Scopes applies to Floors by reference and is not redefined here:

- **Scope Entry** (account-bound, repository-verified, Hash-Pinned Consent, identity lock; entry surfaces hold zero authority) — Annex XIV §XIV.2, §XIV.3, §XIV.6.1; Annex IX §IX.3 unchanged;
- **Scope Graph, Scope Manifest, ratchet inheritance, intersection under multi-membership, change classes** — Annex XIV §XIV.4;
- **Selection-Scoped Context Loading, declared Loading Strategies (lazy/preload/hybrid), index locality, default-silent read path, declared-only telemetry** — Annex XIV §XIV.5;
- **Publisher Orchestrator, static/live boundary, session adoption, epochs, leadership arbitration** — Annex XIV §XIV.3.2, §XIV.6;
- **Identity classes ID0–ID3 and per-capability Identity Floors** — Annex XIV §XIV.7;
- **Governed local actuation through Execution Providers under Capability Mapping** — Annex XIV §XIV.8, itself an instantiation of the Annex IX Finalizer lifecycle.

What this annex adds is the **spatial machinery** Annex XIV deliberately leaves open: the combined entry capture and spatial-registration procedure; the Virtual Replica bridge to the main paper; the sight-field resolution model behind headless designation; the **Spatial Anchor Registry and its target-descriptor records**, filling the extension point declared in Annex XIV §XIV.8.2 and §XIV.12 for spatially anchored capabilities; the Pose Authority for mobile objects; identity facets with visible confirmation; geofencing as a context layer; and the preparation-view binding.

### XV.1.3 Relationship to Annex XI — Designation Instruments (Normative)

Designation uses the WR Pointer unchanged. Per the discipline of Annex XIV §XIV.9, this annex adds **no designation semantics and no instrument classes of its own**; genuinely new instrument classes are added in Annex XI under its §XI.20. Annex XI v1.3 already carries identifier scan, proximity designation, terminal/industrial input, declared WR Link activation, and hand-ray/spatial-gesture designation in the live visual field (§XI.4.1), with the Unmapped Target Fallback (§XI.5) and explicit context bounding (I8).

This annex **requires and declares as a companion Annex XI §XI.20 amendment** three instrument/source-class additions: **sight-field recognition designation** (AI-vision candidate recognition within the active camera field of view), **named addressing** (spoken registered identity name), and **gaze/head-direction designation**. This annex specifies the spatial resolution machinery behind these instruments (§XV.5–§XV.6, §XV.10–§XV.11); their catalog admission, origin classes, and governance live in Annex XI.

### XV.1.4 Further Cross-References (Normative)

Preparation lifecycle semantics are defined in Annex VI and are referenced, not redefined, here. BEAP™ handshake architecture is defined in Annex VII. Entry-code principles ("the code is an identifier and invitation — not authority"), consent classes C0–C4, and the Finalizer execution lifecycle are defined in Annex IX. Capability-governed information flow and WR Guard™ enforcement are defined in Annex X. PoAC™ governs every capability admission, including local spatial-mapping admissions (§XV.10.2).

### XV.1.5 Worked Example (Informative) — The Combined Entrance Capture

A technician arrives at a production hall. At the door hangs one printed label: an optical WR Code® carrier and an AprilTag side by side on the same sheet. She looks at it through her smart glasses — one glance, one camera frame, both markers detected simultaneously. From that single act, two chains run in parallel. The WR Code® component resolves the Floor: repository reference, signed Scope Manifest, Hash-Pinned Consent at the Scope Envelope, the mandatory WR Handshake with the hall's Upper Group — everything verified against the repository, nothing trusted from the label itself, which holds zero authority. The AprilTag component resolves her position: its family, ID, physical size, and Floor pose are recorded in the pinned Spatial Anchor Registry, so the runtime computes her camera pose in Floor coordinates from the same frame and registers her inside the hall's existing Virtual Replica.

By the time she steps through the door, both halves have converged: she is consented into the Scope *and* localized within it. SLAM takes over tracking; the robots and machines entering her field of view become local candidates; no individual object will need a physical scan. A smartphone visitor gets the identical result over a different carrier pair — NFC tap for the BEAP™ resolution, camera on the same label for the fiducial. What the entrance double scan did **not** do is equally important: it created no Replica (it registered her into one that already existed), it proved no object identity (fiducials are candidates and anchors, never proofs), and it granted no authority beyond the consented Scope Envelope. One glance opened a governed, spatially registered world — and everything after it is designation, not scanning.

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## XV.2 Terminology and Bridging Definitions

Terms defined in Annexes II–XIV apply as defined there. This annex adds:

**Floor.** A Scope (Annex XIV) whose members are physical objects and whose Scope Manifest declares spatial anchoring. In the main paper, this construct appears as the roof WR Code® spatial group; "Floor" is the Canon term. A Floor's **Upper Group** is the WR group whose WR Handshake governs the Floor Scope.

**Virtual Replica.** The Canon term for the 3D environment and object replica that the main paper describes generically as a "digital twin" or "3D replica." The construct is unchanged; the Canon designation is Virtual Replica. A Virtual Replica may originate from LiDAR capture, photogrammetry, BIM or CAD data, smartphone room scanning, manual surveying, or the user-annotation capture workflow described in the main paper.

**Floor Entry.** Scope Entry (Annex XIV §XIV.2) into a Floor, extended by Spatial Registration where the Floor's declared entry profile requires it. Account binding and repository reachability apply per Annex XIV §XIV.3.1 without exception.

**Entry Capture.** The physical act of capturing the Floor's entry information, carrier-neutral. "Scan" is used informally for optical capture only.

**Spatial Registration.** Aligning the device's camera/sensor space with the Floor coordinate space against known anchors.

**Object Designation.** The user-side act of indicating an intended member object — by name, pointing, gaze, head direction, proximity contact, WR Pointer selection, spatial context, or combination — through the instruments of Annex XI §XI.4.1 including the additions of §XV.1.3. In camera-based modes, designation operates on objects that **AI-vision object recognition** has detected in the active camera stream and associated with candidate member identities; recognition is the perceptual substrate on which the instruments act. Designation expresses intent; it carries no authority.

**Identity Resolution.** The runtime-side act of resolving a designation to exactly one WR identity, its Virtual Replica, and its declared capability set, inside the active WR Handshake context.

**Headless WR Code® Scan (Headless Resolution).** Identity resolution performed without a per-object physical capture, inside an established WR Handshake context. Its basis technical concept is **AI-vision object recognition**: the runtime recognizes the physical object in the camera stream and associates it — corroborated by registered fiducials, Verified Associations, and spatial mapping — with the member identity whose WR Code® a physical capture would have resolved. The recognition result takes the place of the physical code capture; the resolved identity, governance, and proof obligations are identical to those of a physical WR Code® capture; only the capture instrument differs.

**Verified Association.** The governed act of associating a designation evidence source (e.g., an NFC contact, a fiducial detection, a recognized visual signature) with a WR member identity through a signed record in the Spatial Anchor Registry. The term "pairing" is not used for object selection; proximity-based designation is a Verified Association lookup, not a security pairing.

**Spatial Anchor Registry.** The versioned, signed registry of a Floor's spatial anchors and target-descriptor records (fiducial records, mapped positions, Verified Associations), produced under the Annex VII artifact machinery, pinned by digest from the Floor's Scope Manifest, and filling the target-descriptor extension point of Annex XIV §XIV.8.2/§XIV.12.

**Recognition Package.** The entry-class recognition substrate of a spatially anchored Floor: the declared AI-vision adapter(s) for the Floor's member population together with per-member recognition descriptors, declared in the Scope Manifest, hash-verified, and loaded at Floor Entry before sight-field candidate formation begins (§XV.5.6).

**Pose Authority.** The party declared in the Scope Manifest as authorized to sign dynamic pose updates for a mobile member object.

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## XV.3 Principles

The following principles are normative for every implementation profile of this annex.

**P1 — Floor Entry is Scope Entry.** Floor Entry always includes the WR Handshake of Scope Entry per Annex IX §IX.3 and Annex XIV — account-bound, repository-verified, consented at the Scope Envelope. No designation mode, carrier convenience, or geofencing signal replaces or weakens it. Spatial Registration extends entry; it never substitutes for it.

**P2 — Transport agnosticism of the WR Code®.** The WR Code® is based on BEAP™ and is not synonymous with any carrier. QR is one possible optical carrier; NFC, radio mechanisms, and other physical or digital carriers may transport the same logical identity and resolution information. The transport mechanism is separate from the logical WR identity and resolution model.

**P3 — Designation is never authority.** Consistent with Annex IX §IX.3.1 and the zero-authority entry surfaces of Annex XIV §XIV.6.1, a designation — a spoken name, a pointing gesture, a recognized object, a detected fiducial, a proximity contact — is evidence of intent and a resolution key. It is never a cryptographic proof and never an authorization. Sensor evidence selects; BEAP™ and the WR Handshake prove; consent and Finalizers authorize.

**P4 — Sight-Field Requirement.** In sight-field designation mode, the target object must be inside the active camera field of view and recognized before it can be designated through that mode. Recognition begins when an object enters the camera stream; candidate association to WR member identities is continuous, passive, and local.

**P5 — Modality freedom.** There is no mandatory modality sequence (no "first point, then speak"). Designation may use one instrument or any combination; the runtime decides evidence sufficiency per P8. The system supports interaction profiles rather than one universal selection method.

**P6 — Visible identity confirmation.** The resolved identity is visibly displayed to the user upon resolution — manufacturer, model, type, and/or instance identity as appropriate to the bound facet (§XV.8). No consequential action proceeds on an unconfirmed or ambiguous resolution.

**P7 — Instrument invariance.** A Headless Resolution is semantically identical to a physical WR Code® capture of the same member: same resolution path, same Selection-Scoped Context Loading (Annex XIV §XIV.5.1), same governance. Capability admission, consent classes, logging, and proof

obligations do not vary with the designation instrument or device class (Annex XI I9). Instrument availability is a hardware fact, not a device-class property.

**P8 — Consequence-scaled sufficiency and verification.** Evidence sufficiency thresholds are not free-standing values; they are bound to the consequence class of the intended interaction. Read-only display may accept lower designation confidence with visible confirmation; consequential invocations require multi-evidence designation or explicit confirmation regardless of confidence, and run at the declared consent-class floor C0–C4 and Identity Floor of the capability schema (Annex XIV §XIV.8.2), escalating under the ratchet. Designation confidence ranks candidates; it never authorizes.

**P9 — Signed spatial mappings.** Every association between a physical evidence source and a WR member identity exists only as a signed record in the Spatial Anchor Registry, which is versioned separately from the Scope Manifest, pinned by digest from it, monotonic, freshness-bound, and rollback-protected under the Annex IX manifest discipline. Registry updates never occur as silent replacement.

**P10 — Geofencing is context, never proof.** Geofencing signals inform preselection, preloading, zone policy, and revocation. They never replace BEAP™ resolution, Scope Entry, or precise spatial registration, and never serve as cryptographic evidence. Geofence evaluation is local and data-minimizing by default; it produces no publisher-visible signal unless declared as telemetry under Annex XIV §XIV.5.4, and persistent movement histories are not required to establish current zone presence.

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## XV.4 WR Code® Foundations, Compact Payload, and Loading

The WR Code® used for Floor Entry does not itself carry the Floor, environment, or Virtual Replica data — that data lives in the repository and is resolved from there. The code is an identifier and invitation, never authority (Annex IX §IX.3.1). It contains compact identity, integrity, and repository-resolution information. Representative compact fields:

- `publisher_id`
- `floor_id` (the Scope identifier)
- `repository_reference`
- `manifest_id`
- `expected_manifest_hash`
- `code_anchor_id`
- `signature`
- `protocol_version`

The Scope Manifest, Scope index, Spatial Anchor Registry, member contexts, and Virtual Replica resources are loaded from the repository after verification. Loading follows Annex XIV §XIV.5 unchanged: the Scope index loads at entry; member contexts and the execution artifacts their declared capabilities reference load **upon designation**, hash-verified against the Scope Manifest; the declared Loading Strategy (lazy default; preload/hybrid for constrained-connectivity Floors)

governs any entry-time preloading. The Virtual Replica, the Spatial Anchor Registry, and the Recognition Package (§XV.5.6) are entry-loaded resources of a spatially anchored Floor profile; per-member Replica detail may be declaration-bound to designation.

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## XV.5 Floor Entry and the Combined Entry Capture

### XV.5.1 Floor Entry as one logical operation (Normative)

Floor Entry is defined as **one logical operation** with invariant steps, executed over interchangeable transport profiles:

1. Entry Capture resolves the Floor reference (BEAP™/WR Code® payload, any carrier).
2. Scope Entry is performed per Annex IX §IX.3 and Annex XIV: verification chain, Hash-Pinned Consent at the Scope Envelope, identity lock; default identity class ID0 unless the Floor declares a higher entry floor (e.g., ID2 for an industrial Floor).
3. The signed Scope Manifest is loaded and verified against `expected_manifest_hash`; the Spatial Anchor Registry version pinned in the manifest is loaded and verified.
4. The Virtual Replica and the Recognition Package (§XV.5.6) are loaded per the declared Loading Strategy (or located in cache and freshness-checked under the Annex IX staleness rules).
5. Spatial Registration is performed against one or more registered anchors where the Floor's entry profile declares it.
6. The runtime activates the Floor context: the Scope index makes member identities, aliases, spatial anchors, and permitted relationships addressable; sight-field candidate recognition begins once the Recognition Package is verified.

After successful Floor Entry, individual member objects normally do not need to be captured again. The physical WR Code® retains four roles: initial entry mechanism, recovery mechanism, fallback, and direct verification mechanism where a capability schema requires renewed direct verification (§XV.10.3).

### XV.5.2 Combined entry capture (Normative)

The entry point may present a **combined capture surface** so that steps 1 and 5 begin in a single user act:

- **Combined optical label:** one printed label containing both an optical BEAP™/WR Code® carrier and an AprilTag (or equivalent fiducial). Both markers are detectable in the same camera frame. The BEAP™/WR Code® component identifies and resolves the Floor Scope, repository, and appropriate Virtual Replica; the fiducial component provides a known spatial anchor and camera-pose reference.
- **NFC + fiducial:** on smartphones, NFC-based BEAP™ resolution combined with simultaneous fiducial camera detection produces the same user experience over a different carrier pair.

The combined capture is a **transport profile** of the single logical Floor Entry operation, consistent with P2. The entry surface carrying the combined label holds zero authority (Annex XIV §XIV.6.1);

everything verifies against the repository. Future carriers may be added as further profiles without altering the operation's invariant steps.

### XV.5.3 Spatial Registration (Normative)

The fiducial anchor has a known pose in the Floor coordinate system, recorded in the Spatial Anchor Registry together with its family, ID, and physical size. The camera computes the anchor pose relative to itself. The device pose in Floor coordinates is obtained approximately as:

$$T_{\text{floor\_camera}} = T_{\text{floor\_tag}} \times \text{inverse}(T_{\text{camera\_tag}})$$

A single anchor may be sufficient for initial registration when its physical size is known and registered, its Floor pose is registered, the camera is calibrated, and the anchor is clearly visible. For higher reliability, an entrance may register multiple spatially separated anchors; multiple anchors permit detection of incorrect mappings and reduce orientation and pose error.

### XV.5.4 Registration, not creation (Normative)

The entry capture does not create the Virtual Replica. It **registers and relocalizes** the device within an already existing Replica. Replica provenance (LiDAR, photogrammetry, BIM/CAD, smartphone room scanning, manual surveying, annotated capture workflows) is recorded in the Scope Manifest; the entry operation is provenance-agnostic.

### XV.5.5 Continuous tracking (Normative)

After initial registration, SLAM or visual-inertial odometry tracks the device continuously. Registered fiducials and other spatial anchors periodically correct accumulated drift. The combined entrance process is therefore characterized as **bootstrapping, registration, and relocalization of the device within the existing Virtual Replica**.

### XV.5.6 The Recognition Package (Normative)

Reliable sight-field designation requires that the runtime can recognize the Floor's members **before** any of them is selected. The recognition substrate is therefore entry-class, never selection-class:

1. The Scope Manifest of a spatially anchored Floor declares a **Recognition Package**: the AI-vision adapter(s) for the Floor's member population — e.g., publisher-supplied LoRA adapters, delivered as repository artifacts under the Canon's artifact classes and hash-identified — together with per-member **recognition descriptors**: compact visual signatures sufficient for detection and candidate association, plus fiducial and Verified-Association references into the Spatial Anchor Registry.
2. The Recognition Package is loaded and hash-verified at Floor Entry, as part of the entry-loaded material, before sight-field candidate formation begins. Adapter admission follows the Canon's adapter discipline — PoAC™-governed, publisher-attested, hash-verified — the same machinery that governs Pointer/Overlay adapters in Annex XI.
3. For very large Floors, the manifest MAY partition recognition descriptors along child-Scope boundaries (**zone packages**) under the declared Loading Strategy; a zone's package is loaded no later than zone entry, and geofence-assisted preloading (§XV.12) may pull it earlier. Partitioning bounds package size; it never defers recognition below the granularity of the zone the user is in.



4. **Distinction from Selection-Scoped Context Loading.** The Recognition Package answers *"what is visible, and which member might it be"* and is entry-class; member contexts, execution artifacts, and detailed Replica resources answer *"what does this member offer"* and remain selection-loaded per Annex XIV §XIV.5.1. Recognition never requires selection; selection never precedes recognition.
  5. Real-time, per-object retrieval of recognition data is **non-conforming as the primary sight-field path**: it inverts the dependency, since an object cannot be recognized with data that only its selection would load, and it couples recognition reliability to per-object network latency. Runtime retrieval is permitted only as declared refinement — e.g., loading higher-fidelity verification material for an already-designated member — and for non-visual designation paths (named addressing, proximity designation), which do not depend on visual recognition.
  6. Recognition executes locally on the entry-loaded package; recognition queries leave neither the device nor the session, consistent with the local read path and telemetry discipline of Annex XIV §XIV.5.3–5.4.
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## XV.6 Headless Object Designation

### XV.6.1 Definition (Normative)

Headless Object Designation is the resolution of an individual member's WR identity **without a per-object physical capture**, performed entirely inside the established WR Handshake context of the active Floor. It is normatively a **headless WR Code® scan**: the designation resolves exactly the identity that a physical capture of that member's WR Code® would resolve, through the same resolution path, followed by the same Selection-Scoped Context Loading (Annex XIV §XIV.5.1), with identical governance. Under P7, no property of trust, capability, consent, or logging depends on whether resolution was triggered by a physical capture or a headless designation.

An object recognized in the sight field for which the Scope Manifest declares no member context is an **unmapped target**; the Unmapped Target Fallback of Annex XI §XI.5 applies verbatim, including its no-publisher-signal rule.

### XV.6.2 Sight-field mode (Normative)

In sight-field mode, recognition runs continuously on the active camera stream (Smart Glasses, smartphone, or other camera-bearing device), executing locally on the entry-loaded Recognition Package (§XV.5.6). As objects enter the field of view, the runtime associates them locally with candidate member identities using the package's AI-vision adapters and recognition descriptors, registered fiducials, spatial mapping, Verified Associations, and any other registered evidence source. The sight-field requirement (P4) is the mode's defining constraint: an object outside the active field of view cannot be designated through this mode (it may still be designated through non-visual instruments where the profile permits, e.g., unambiguous named addressing under §XV.6.4).

The user does not necessarily have to point. The user does not necessarily have to speak. Candidate formation is passive and local; per Annex XIV §XIV.5.3–5.4, it produces no publisher-side signal. Designation occurs when the instrument evidence crosses the sufficiency bound of P8.

### XV.6.3 Designation instruments (Normative, by Reference)

Designation may use one instrument or a combination, drawn from the Annex XI §XI.4.1 catalog: hand-ray/spatial-gesture designation in the live visual field, identifier scan, proximity designation, terminal/industrial input, cursor and touch — plus the three additions this annex declares under Annex XI §XI.20 (§XV.1.3): sight-field recognition designation, named addressing, and gaze/head-direction designation.

All instruments inherit Annex XI governance unchanged: offer-never-action, content-never-instruction, signed origin classes, explicit context bounding (I8). Spoken designation commands (e.g., a command of the "Show Details" class bound to the currently focal member) are **declared vocabulary in the capability catalog**, never free-form instruction parsing at the trust boundary; the command class and its binding semantics are declared in the Scope Manifest.

### XV.6.4 Named addressing and co-worker interaction (Normative)

Every member may carry a registered human-readable name (e.g., Uwe065), recorded in the Scope index with its Verified Associations. Named addressing designates by name alone when the runtime already holds a reliable association between the called identity and a physical instance — through spatial mapping, visible markers, fiducials, an attested pose, or prior designation.

For robots and autonomous systems, this enables interaction modeled on addressing a human co-worker: the user looks at the robot (satisfying the sight-field requirement where the profile demands it), speaks its name and an instruction — "*Uwe065, get the wrench*" — and the runtime resolves Uwe065 to its member identity and loads exactly that unit's catalog entry: declared context, declared command set, and, by reference, the automation authorizations behind them (Annex XIV §XIV.5.1). The spoken instruction is input to the AI-as-router model of the Canon: the AI selects among the pre-declared chains of the signed capability catalog; it never synthesizes new paths. Invocation runs the unchanged path — client-generated preview, consent at the declared class and Identity Floor (consequential robot commands typically carry ID2 floors and C2/C3 consent on an industrial Floor), Execution Provider as Finalizer, witnessed transition, sealed receipt (Annex XIV §XIV.8).

When the called identity remains ambiguous (multiple plausible instances), the runtime requests or infers an additional signal — pointing, gaze, proximity, closer inspection — or offers candidate cycling per §XV.6.5.

### XV.6.5 Ambiguity handling (Normative)

Under ambiguity, the runtime never auto-selects when an invocation would follow. Candidates are ranked and visibly highlighted; the runtime offers the **cheapest available disambiguation instrument** for the active profile (gaze dwell, head-movement cycling, a confirmation word, a proximity tap). For pure information display, the top candidate may be shown provisionally, visibly marked as provisional. Candidate cycling and highlighting are available hands-free where the device supports it.

### XV.6.6 Designation on the live Virtual Replica (Normative)

Designation is not restricted to physical co-presence. In an operator room, the live Virtual Replica of a Floor renders mobile members — robots and autonomous systems — at their attested poses

(§XV.11). The WR Pointer designates an individual unit **on the Replica** exactly as it would in the real sight field: same designation semantics, same resolution path, same Selection-Scoped Context Loading, same governance (P7). Pointer-on-Replica and Smart-Glasses-in-situ are two embodiments of one designation model; only the field of view differs. The Replica view continues or reopens the same standing Scope session at the workstation; it is a continuation of the same governed context, never a new session.

Live pose rendering is a live-class capability under the static/live boundary of Annex XIV §XIV.3.2 unless poses are delivered through declared slots: with the Publisher Orchestrator offline, the Replica degrades visibly to last-verified poses, marked stale, and designation-for-invocation on stale-posed mobile members is suspended pending freshness (§XV.11).

### **XV.6.7 Device embodiments (Informative)**

The concept provides its strongest benefit with Smart Glasses, which naturally combine continuous camera view, field-of-view awareness, gaze or head direction, speech, pointing, spatial tracking, and direct visual identity overlays, enabling largely hands-free and continuous interaction. The mechanism is not restricted to Smart Glasses: a smartphone camera performs visual recognition, fiducial detection, and field-of-view designation; the experience is less continuous and less hands-free but technically complete. Per Annex XI I9, device class shapes ergonomics, never semantics.

A smartphone user does not scan a WR Code® for every individual product after Floor Entry. NFC at a product, shelf, or price label is a proximity designation (Annex XI §XI.4.1) resolved through a Verified Association inside the active Floor context; this is object designation, not a replacement for Scope Entry (P1) and not a security pairing (§XV.2).

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## **XV.7 Interaction Profiles**

The same system supports different interaction profiles rather than enforcing one universal selection method. Representative profiles:

- **Quiet retail environment.** Speaking aloud is undesirable; pointing, gaze, WR Pointer, or proximity designation are preferred. The retail flow is the electronics-store Scope of Annex XIV §XIV.1.2, extended by this annex's sight-field machinery: one-time Floor Entry at ID0 → user approaches a product → NFC tap at the price label or sight-field designation → runtime resolves the member inside the active Floor context → product context, Virtual Replica detail, and declared capabilities load.
- **Loud industrial environment.** Speech recognition is unreliable; pointing, WR Pointer, visible markings, or technical localization are preferred.
- **Hands-busy environment.** Name-only or gaze-based designation is preferred.
- **Many nearby similar objects.** The runtime combines identity names, spatial mapping, pointing, gaze, and visible markers; ambiguity handling (§XV.6.5) is routinely active.
- **Widely separated objects.** A spoken name or coarse direction is already sufficient.
- **Identical units with distinct identities.** Five visually identical robots each carry a unique member identity; name-only designation works when the runtime holds a reliable instance association; otherwise an additional signal is requested.

Profiles are declared per Floor or per zone (child Scope, §XV.12) and select the instrument set, sufficiency policy, and disambiguation defaults; child Scopes may only tighten (Annex XIV §XIV.4.3), and profiles never modify governance (P7).

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## XV.8 Identity Model and Visible Confirmation

The system is **identity-centric, not category-centric**. Machines, robots, industrial equipment, tools, consumer products, shelves, product packages, files and document folders, and other uniquely addressable physical entities can all be WR members.

A WR member identity exposes orthogonal **facets**:

- **instance identity** — this physical unit;
- **type identity** — this product/model class;
- **batch identity** — this production batch;
- **position identity** — this shelf or mapped location.

Ten identical product packages share one type identity while carrying ten distinct instance identities. A capability declares which facet it binds, carried in its target descriptor (§XV.10.2; Annex XIV §XIV.8.2): a price comparison binds the type facet; a maintenance order binds the instance facet. Declared facet binding keeps identical-unit scenarios out of the security discussion entirely: capability invocation binds to the declared facet, and resolution to a different facet is a mis-specification, not a permissible substitution.

**Visible confirmation (P6).** On resolution, the runtime visibly displays the resolved identity — for a product, e.g., manufacturer, exact model, product type, and, where the bound facet requires it, the individual identity — and may visually highlight the selected object in the Smart Glasses or smartphone display. Visible confirmation prevents the user from unknowingly acting on the wrong object and is a precondition for consequential invocation. Designation alone never identifies the person: selection is member resolution at the session's identity class, not identity disclosure (Annex XIV §XIV.7).

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## XV.9 Identification Provisioning Spectrum (Informative)

Identification methods vary with cost, operational complexity, and safety requirements; the designation model is provisioning-agnostic.

- **Low-cost provisioning:** physically displayed human-readable names; color coding or other visible markings (especially useful when similar robots or machines operate near one another); printed fiducials; NFC tags; fixed placement.
- **Professional provisioning:** detailed spatial maps, central control-system integration, continuously attested poses, UWB or equivalent localization, cryptographic device identities.

Manufacturers and operators may combine visible names, color coding, fiducials, NFC, Bluetooth/beacon mechanisms, optical WR Code® carriers, and fixed spatial mapping in any

mixture. The provisioning mixture changes recognition reliability and disambiguation cost; it never changes governance.

---

## XV.10 Security Model

### XV.10.1 Evidence taxonomy (Normative)

Object recognition, pointing, gaze, and fiducial detection do not by themselves prove a cryptographic identity. A fiducial can be copied, photographed, moved, or replaced. Physical evidence sources are therefore classified strictly as:

- **visual candidate identifiers** (recognition, fiducial ID),
- **spatial anchors** (pose references),
- **references into signed mappings** (Verified Associations).

The cryptographic proofs of the system are carried exclusively by BEAP™ and the WR Handshake; no separate proof system is introduced by physical designation (P3).

### XV.10.2 The Spatial Anchor Registry (Normative)

A signed target-descriptor record in the Spatial Anchor Registry associates, per anchor:

fiducial family + fiducial ID + known physical size  
+ Floor pose  
+ WR member identity (and bound facet)  
+ Virtual Replica reference

Equivalent record classes exist for NFC/proximity tags and registered visual signatures. During installation, an operator resolves the member's WR identity, captures or assigns the anchor, captures the pose, associates the records, signs the mapping, and publishes it through the repository.

**Creating or changing a mapping is a PoAC™ admission event** — evidence-backed, reviewed, signed, revocable, proposed and never silently applied — mirroring the Capability Mapping discipline of Annex XIV §XIV.8.3(2) on the designation side. The registry is versioned separately from the Scope Manifest, pinned by digest from it (P9), monotonic, freshness-bound, and rollback-protected.

The Spatial Anchor Registry is the concrete target-descriptor machinery for spatially anchored capabilities reserved as an extension point in Annex XIV §XIV.8.2 and §XIV.12.

**Blockchain anchoring (Normative).** Registry versions and individual target-descriptor records MAY additionally be anchored on a public blockchain under the Canon's tiered evidence model (Annex IX, tiers L/C/X — e.g., Solana anchoring). The main paper already applies such anchoring to WR Code® artifacts and signed contributions; this annex extends the same discipline to the **locality records**: the fiducial-to-identity-to-pose mappings themselves. An anchored registry version makes tampering with a Floor's spatial truth — a moved, replaced, or re-mapped fiducial published as a silent registry change — externally detectable and strengthens rollback protection beyond the manifest pin. Anchoring is evidentiary only: it proves what was declared and when, and confers no authority (P3).

### XV.10.3 Consequence-scaled verification (Normative)

Critical invocations require verification beyond visual designation. The capability schema (Annex XIV §XIV.8.2) declares the consent-class floor, Identity Floor, and Transition Class; physical-world effects are typically `observed_effect` with its stricter consent-minimum policy. Depending on the declared profile, additional verification includes: a device-level WR Handshake with the object itself (the object cryptographically proves that it is the resolved identity, not merely that something stands at its position), user authorization, a cryptographic NFC device, a hardware key, an additional confirmation, or a renewed direct WR Code® verification. Escalation follows the ratchet discipline: policy and operator may raise the required verification above the declared floor, never below it. Execution itself runs exclusively through the Annex XIV §XIV.8 Execution Provider path under the unchanged five-phase Finalizer lifecycle.

### XV.10.4 Terminology precision (Normative)

NFC used to designate a product is a Verified Association lookup and is not "pairing" in the security sense. The annex vocabulary (§XV.2) — Entry Capture, Spatial Registration, Object Designation, Identity Resolution, Verified Association — is normative; "scan," "pairing," "registration," and "selection" are used only in their defined or informal-explanatory senses.

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## XV.11 Mobile Objects and the Pose Authority

For moving machines and robots, the identity mapping is stable while the spatial pose changes. The two are strictly separated:

- **Static, signed:** the identity binding (anchor/name ↔ WR member identity ↔ Replica reference) in the Spatial Anchor Registry. It never contains a live pose.
- **Dynamic, attested:** the current pose, delivered as a signed feed or as short-lived pose attestations by the Floor's declared **Pose Authority** (e.g., the control system). The registry declares *who may attest poses* for a member; it does not embed the poses.

Pose delivery sorts into the static/live boundary of Annex XIV §XIV.3.2 by declaration: slot-delivered pose snapshots are static-class; continuous pose streams are live-class and require the adopted Publisher Orchestrator. Pose freshness requirements scale with assurance class: information display may accept cached poses, visibly marked stale under the Annex IX staleness rules; designation-for-invocation on a mobile member requires pose attestations within the freshness bound declared for the consent class. The live Virtual Replica in the operator room (§XV.6.6) renders Pose-Authority-attested poses; designation on stale poses is downgraded to provisional per §XV.6.5.

This separation updates mobile-member positions at arbitrary frequency without re-signing identity mappings and without weakening them.

---

## XV.12 Geofencing as a Context and Policy Layer

Geofencing answers "*which site, building, Floor, or zone is the device probably inside?*" It cannot normally prove the exact object or exact indoor position, and per P10 it never substitutes for BEAP™ resolution, Scope Entry, or precise spatial registration.

Devices may evaluate geofences using GNSS/GPS, cellular information, Wi-Fi, BLE beacons, UWB, camera localization, or combinations. A practical evidence hierarchy:

|                         |                                      |
|-------------------------|--------------------------------------|
| GPS / cellular          | → likely site                        |
| Wi-Fi / BLE             | → likely building, Floor, department |
| Fiducial / SLAM / UWB   | → precise position and orientation   |
| Proximity / designation | → specific member object             |

NFC is not conventional geofencing: it confirms a very close contact point, not membership in a broad area.

Permitted geofencing functions: preselecting likely Floors; preloading Scope Manifests or Replicas per the declared Loading Strategy; activating Floor- or zone-specific interaction profiles and rules (zones as child Scopes under ratchet inheritance); distinguishing public, staff, and safety zones; reducing candidate ambiguity; detecting Floor exit; revoking temporary permissions; requiring a new Floor Entry. Geofencing outputs are probability and context signals evaluated locally; they reach no publisher and no orchestrator unless declared as telemetry under Annex XIV §XIV.5.4. Establishing current zone presence requires no persistent movement history.

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## XV.13 Designation as Gateway to the Object Information Space

Designating a member immediately makes its governed information space available through Selection-Scoped Context Loading: current state (through declared slots), operating capabilities (the context-filtered catalog view), available commands, documentation, maintenance history, safety information, current tasks, and preparation briefings — exactly what the declaration covers and the session's identity class and permissions admit. Designation thereby connects the physical object directly with its normalized digital context — the Virtual Replica and everything the active WR Handshake context authorizes for this user, this facet, and this zone.

## XV.14 Preparation Views

### XV.14.1 Relationship to Annex VI (Normative)

WR Advance Preparation and WR Predictive Preparation are Canon constructs decomposed from the Forward Preparation lifecycle and specified in Annex VI. This annex does not redefine them; it specifies their **binding to object designation**: how they are triggered by, scoped to, and displayed from a designated physical member.

### XV.14.2 WR Advance Preparation binding (Normative)

Advance Preparation is triggered by a known upcoming event, task, or operation — planned maintenance, a major machine disassembly, loading a machine onto a truck, coordinated airport

operations, or another complex multi-person, multi-step event. The AI prepares the available information before the event into a highly structured, prioritized, visually consumable briefing, which may include maintenance schedules, intervals, upcoming actions, dependencies, involved people and responsibilities, required tools or resources, calendar information, historical failures, known problems, warnings, risks, recommendations, and suggested priorities.

Bound to designation, the flow is: user designates the machine (Smart Glasses, Pointer, or Replica) → identity resolves → member context and Virtual Replica detail load → maintenance context becomes available → the Advance Preparation briefing for the scheduled event opens.

### XV.14.3 WR Predictive Preparation binding (Normative)

Predictive Preparation proactively surfaces information likely to be useful, examining current context, available data, historical events, known problems, object state, previous interactions, and relevant memory; it identifies likely risks, overlooked dependencies, probable failure points, recommended actions, unusual patterns, or information the user may need soon. It is grounded in available facts and context; its inferences are clearly identifiable as such.

### XV.14.4 Complementarity (Normative)

Advance and Predictive Preparation are complementary views, not rigidly isolated systems. The primary distinction is trigger and emphasis: Advance prepares a known event; Predictive proactively surfaces likely relevance. An Advance briefing may contain a Predictive section; a Predictive briefing may later become part of a formal Advance process once an event is scheduled. No artificial hard boundary is imposed.

### XV.14.5 Rendering, typing, and permissions (Normative)

Preparation views render as normalized, visually structured pages (generated-web-view class) associated with a WR member, a Floor, a machine, an object group, a planned event, or an operational process. Views are generated locally or within a controlled environment; **all inputs are normalized before presentation**, and the output is placed in or referenced through the repository. The mechanism never depends on passing arbitrary, unverified attachments directly to the runtime.

Content blocks are **typed and machine-readable**: *fact* (with provenance, PoAC™-referenced), *recommendation*, and *prediction* (with confidence indication). Predictions inform; they never authorize — a Predictive warning cannot itself unlock an action (ratchet discipline).

Permissions are inherited by **intersection, not union** — consistent with the multi-membership rule of Annex XIV §XIV.4.3: a preparation view's content is bounded by  $\text{permissions}(\text{member}) \cap \text{permissions}(\text{event}) \cap \text{permissions}(\text{operator})$ , and the view is generated filtered per recipient — never as a full view with client-side hiding.

Advance briefings are cacheable as signed repository artifacts bound to their event, with declared freshness bounds; Predictive content is generated fresh from normalized inputs.

## XV.15 Consolidated Operating Flow (Informative)

User approaches the Floor

- geofencing may identify the likely environment (local context only)
- Entry Capture resolves the Floor Scope (BEAP™/WR Code®, any carrier)



- Scope Entry: verification chain, Hash-Pinned Consent, identity lock (ID0 default or declared entry floor)
- signed Scope Manifest + pinned Spatial Anchor Registry verified
- Virtual Replica loaded per declared Loading Strategy
- fiducial anchor establishes initial Spatial Registration
- SLAM/VIO tracks the device; anchors correct drift
- members in the camera field of view become local candidates
- user designates: name, spoken command, pointing, gaze, Pointer, proximity, or combination – no fixed sequence
- runtime resolves and visibly confirms the member identity and bound facet
- member context and referenced execution artifacts load, hash-verified (Selection-Scoped Context Loading)
- Advance or Predictive Preparation view opens on request or by profile
- consequential invocations run at declared consent-class and Identity Floors through the Execution Provider Finalizer path, witnessed and receipted; unmapped recognitions fall to the Annex XI Unmapped Target Fallback with no publisher signal

## XV.16 Deployment Profiles (Informative)

### XV.16.1 Minimum viable retail profile

One combined printed entry label (optical WR Code® carrier + AprilTag or equivalent fiducial); NFC tags at shelves or price labels; type and position facets instead of instance facets; ID0 throughout the read path; no continuous SLAM required — sight-field recognition plus proximity tap suffices; visible confirmation on every resolution. Deployment cost is essentially printing and tagging.

### XV.16.2 High-assurance industrial profile

ID2 entry floor; multiple spatially separated entrance anchors; continuous SLAM/VIO or UWB tracking; Pose Authority feed from the control system; instance facets mandatory; device-level WR Handshake with the object before consequential invocations; `observed_effect` Transition Class with C2/C3 hardware-backed consent per the attestation-adaptive escalation model; full Boundary Event logging per Annex X.

Intermediate profiles are permitted; a profile declares its instrument set, evidence-sufficiency policy, pose-freshness bounds, and verification floors, and may only be escalated, never weakened, by zone policy or operator preference.

## XV.17 Relationship to Prior Interface Classes (Informative)

The designation model is distinguished from adjacent prior interface classes by what each class lacks:

- **Ordinary QR scanning** provides transport without a trust context: a scan resolves a reference, but no verified Scope Entry, no signed manifest, no governed capability catalog, and no consequence-scaled verification follow from it.
- **AR marker tracking** provides pose without identity authority: fiducials anchor graphics, but the marker is not a reference into a signed identity mapping, and detection carries no governance.

- **Pointing and multimodal disambiguation interfaces** provide selection without a cryptographic chain: they resolve *which* object is meant, but resolve it to no proven identity, no trust context, no Virtual Replica, and no governed action path.

Annex XIV §XIV.11 states the claimed contribution of the Scoped Worlds composition; this annex adds the spatial half of that chain — Designation → Identity Resolution → Trust Context → Virtual Replica → Permissions → Action/Preparation — and publishes the combination as prior art under the claims discipline of §XV.0.1.

## XV.18 Extension Points

The following are declared extension points; additions here follow the versioned-amendment rule of §XV.0:

- **E1 — Additional Entry Capture transport profiles** (new carrier combinations for the single Floor Entry operation).
- **E2 — Additional designation instrument/source classes**, added exclusively in Annex XI under its §XI.20 per the Annex XIV §XIV.9 discipline; this annex then specifies their spatial resolution machinery where needed.
- **E3 — Spatial Anchor Registry record classes** beyond fiducial/proximity/visual-signature (e.g., UWB anchors, learned visual landmarks), serving as target-descriptor classes for Annex XIV §XIV.12.
- **E4 — Pose Authority attestation formats** and freshness policies per assurance class.
- **E5 — Preparation-view block types** beyond fact/recommendation/prediction, with declared authorization semantics (informing only).
- **E6 — Profile vocabulary** for declaring interaction profiles per Floor and zone.

## XV.19 Version History

| Version | Date       | Change                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|---------|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.0     | 2026-07-27 | Initial normative specification, with a worked example of the combined entrance capture as informative motivation (§XV.1.5): Floor defined as spatially anchored Scope of Annex XIV, with Scope Entry, Scope Manifest, Selection-Scoped Context Loading, Loading Strategies, telemetry discipline, static/live boundary, identity classes ID0–ID3, and Execution Provider actuation applied by reference; Floor Entry as one logical operation with transport profiles and combined entry capture (optical WR Code® + AprilTag/fiducial; NFC + fiducial); Spatial Registration procedure with multi-anchor reliability model, registration-not-creation discipline, and SLAM/VIO continuous tracking; Recognition Package as entry-class recognition substrate (§XV.5.6: vision adapters and per-member descriptors delivered at Floor Entry, zone partitioning for large Floors, real-time per-object recognition retrieval non-conforming as primary sight-field path); Headless Object Designation grounded in AI-vision object recognition, with sight-field requirement, modality freedom, named addressing and co-worker interaction model, ambiguity handling, and designation on the live Virtual Replica as continuation of the same standing Scope session; designation instruments referenced from Annex XI §XI.4.1 with three additions declared for §XI.20 per the Annex XIV §XIV.9 discipline; unmapped recognitions routed to the Annex XI |

**Version Date****Change**

§XI.5 Unmapped Target Fallback; identity-facet model with visible confirmation; provisioning spectrum; security model with Spatial Anchor Registry as target-descriptor machinery for Annex XIV §XIV.8.2/§XIV.12, mapping creation as PoAC™ admission, optional blockchain anchoring of registry versions and target-descriptor records under the Annex IX tiered evidence model, and consequence-scaled verification; Pose Authority with static/live-sorted pose delivery and visible stale degradation; geofencing as context layer; preparation-view binding to Annex VI constructs; deployment profiles; prior-interface-class differentiation; extension points E1–E6.

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## License Notice

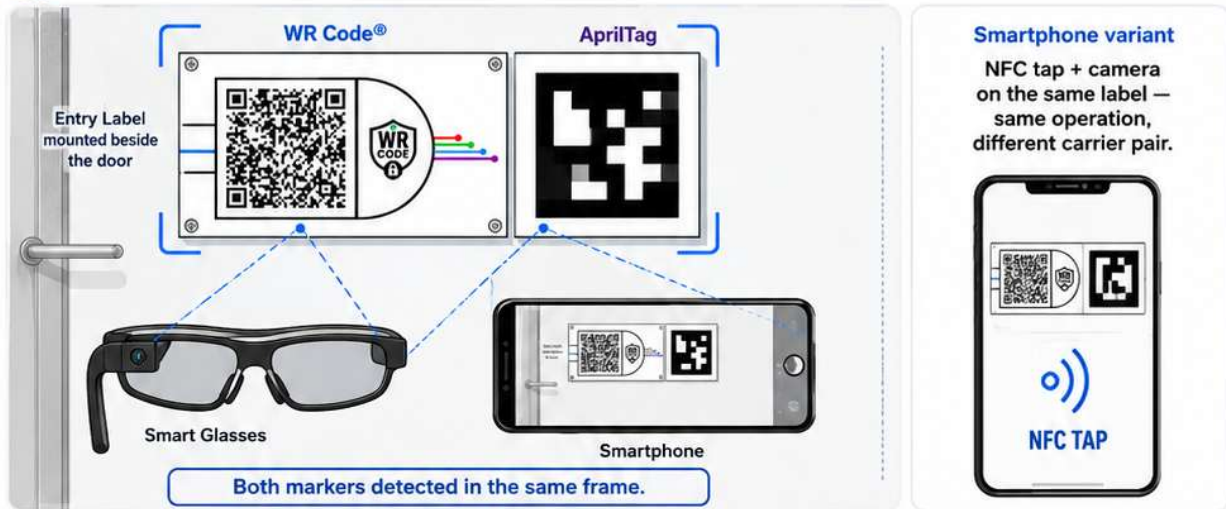
This Annex forms an integral part of the Optirando™ specification and is licensed under the same terms as BEAP™, WR Desk™, and PoAE™.

# The Combined Entrance Capture — One Glance, Two Chains

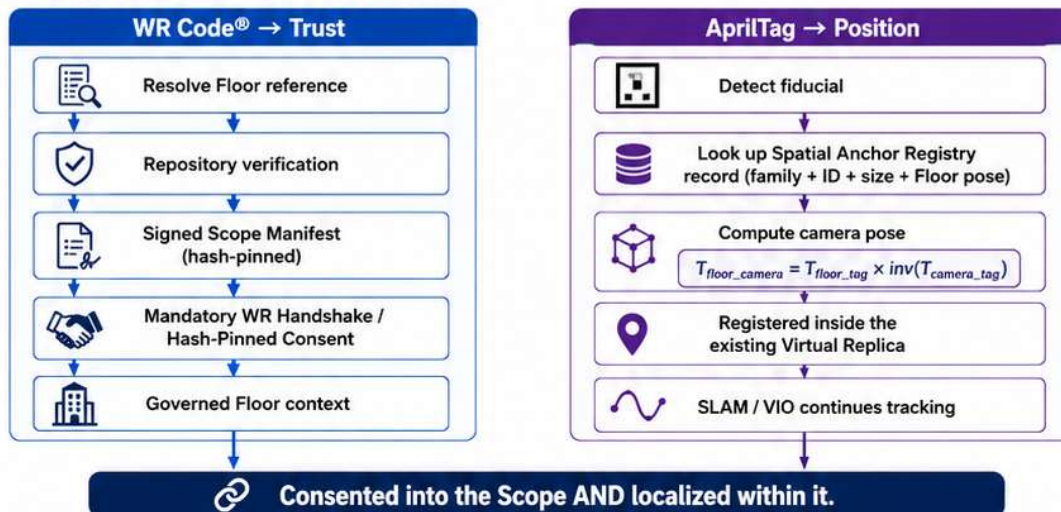
1 / 3

One label, one camera frame: the WR Code® opens the governed Floor, the AprilTag registers you inside it.

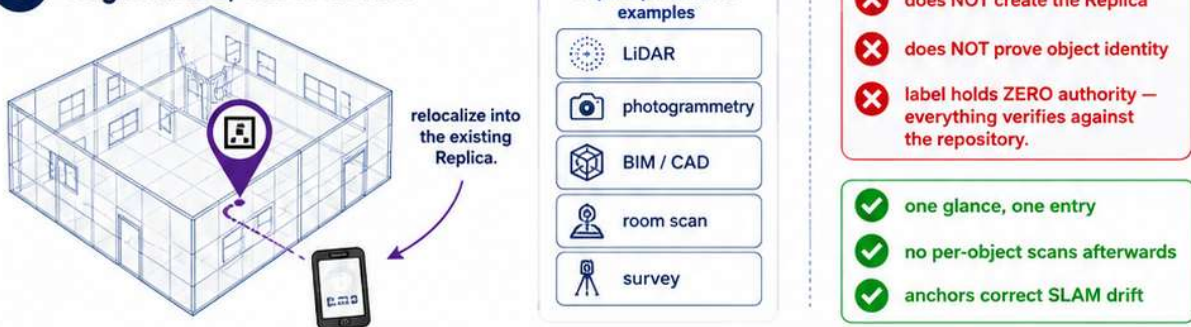
## 1 One label, two markers.



## 2 Two parallel chains from one act.



## 3 Registration, not creation.



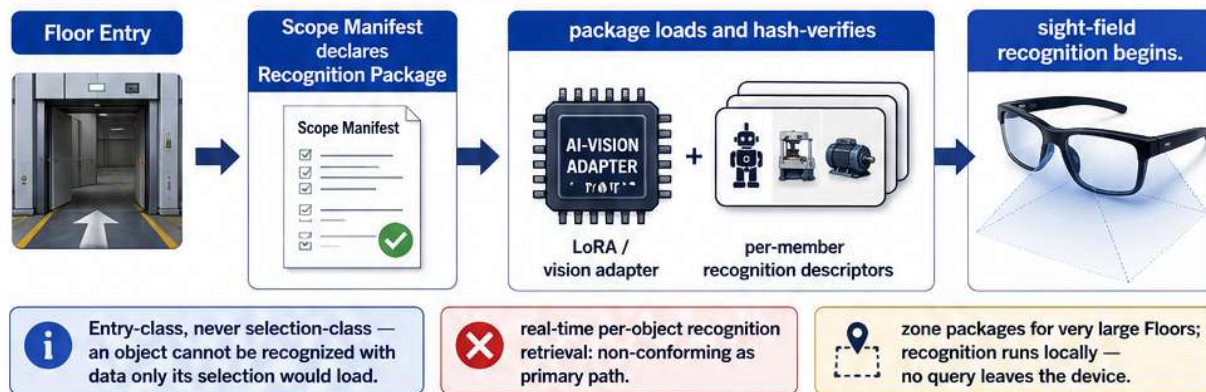
One glance opens a governed, spatially registered world — everything after it is designation, not scanning.



## Infographic 2 — Headless Object Designation — Recognition First, Then Intent

*The recognition result takes the place of the physical code capture — governance identical, instrument different.*

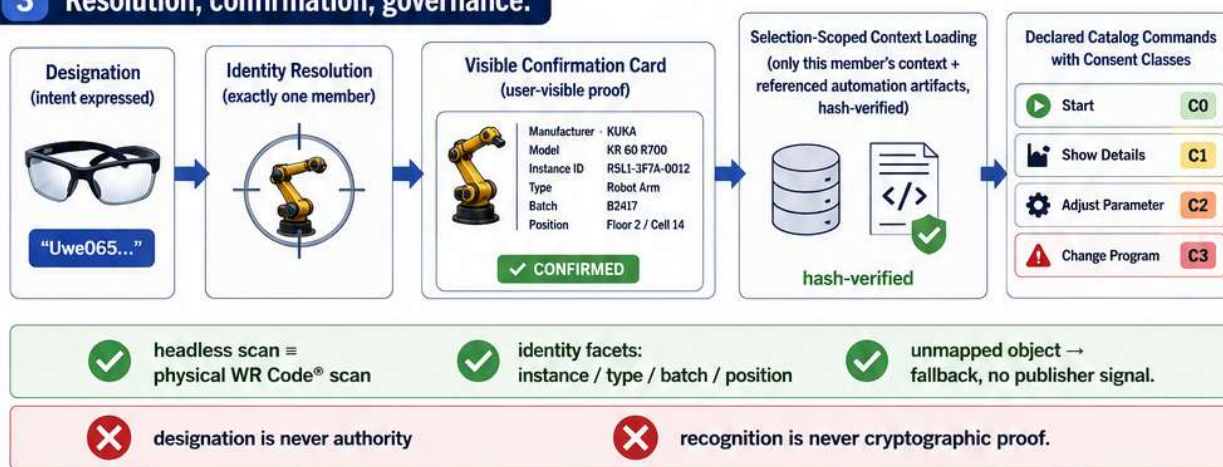
**1 The Recognition Package loads at entry.**



## 2 Sight field + free modality combination.



### 3 Resolution, confirmation, governance.



**Recognition is the perceptual substrate; designation expresses intent;  
only BEAP™ and the WR Handshake prove.**





# Infographic 3 — Spatial Truth, Signed and Anchored

3 / 3

Fiducials are candidates and anchors — never proofs. The mappings are signed, pinned, and blockchain-anchored.

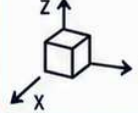
## 1 The evidence taxonomy

### 1. Visual candidate identifier



examples:  
recognition  
result,  
fiducial ID

### 2. Spatial anchor



examples:  
pose  
reference

### 3. Verified Association



examples:  
reference  
into a signed  
mapping

**! an AprilTag can be copied, photographed, moved, replaced.**



Sensor evidence selects · BEAP™ + WR Handshake prove · consent + Finalizers authorize.

## 2 The Spatial Anchor Registry



A. versioned separately,  
pinned by digest from  
the Scope Manifest



B. mapping creation =  
PoAC™ admission —  
proposed, reviewed,  
signed, revocable,  
never silently applied



C. rollback-protected

### Spatial Anchor Registry Record SIGNED RECORD

|                                                                 |                               |                                                       |
|-----------------------------------------------------------------|-------------------------------|-------------------------------------------------------|
|                                                                 | fiducial family + ID          | AprilTag 36h11:23                                     |
|                                                                 | physical size                 | 0.150 m                                               |
|                                                                 | Floor pose                    | x: 2.145 m<br>y: -0.832 m<br>z: 0.000 m<br>yaw: 91.3° |
|                                                                 | WR member identity<br>(facet) | #RSL1                                                 |
|                                                                 | Virtual Replica<br>reference  | VRP://RSL1/AT36h11:23<br>v1.4.7                       |
| record hash<br>7b1d8a44c9e2b70e4a1d2<br>3c7d9e5f6a0b8c2d1e4f7a9 |                               | signed by WR member<br>#RSL1                          |

optional anchoring of  
registry versions and  
target-descriptor records  
(tiers L/C/X, e.g., Solana) —  
tampering with a Floor's  
spatial truth becomes  
externally detectable.

Registry v1.4.7



Public Chain Block  
(e.g., Solana)  
Block #298347821

Evidentiary only: proves  
what was declared and  
when, confers no authority.

## 3 Mobile objects and consequence-scaled verification

### static, signed

Identity Binding (object)



stable identity →  
signed and anchored

### dynamic, attested

Mobile Object (robot)



pose timeline

- ✓ 10:15:03.275
- ✓ 10:15:03.542
- ✓ 10:15:03.809
- ⋮

Pose Authority  
signed by #RSL2



stale pose → visibly stale, invocation suspended.

### Escalation ladder for critical actions

- visible confirmation
- multi-evidence designation
- hardware-backed consent (C2/C3)
- device-level WR Handshake  
with the object itself  
*"the object proves it IS the identity —  
not merely that something stands there"*



identity never re-signed  
for a pose update



ratchet: verification  
may only escalate



governance guarantees enforced  
by Consent Classes and Finalizers



The registry signs the mapping, the manifest pins the registry,  
the chain anchors the truth — and none of it is authority.

